For ACME's new Customer Collaboration platform, the requirements were different again. The product team needed fine-grained access control that supported delegated permissions and dynamic sharing between employees and customers. They chose OpenFGA, a relationship-based authorization system inspired by Google's Zanzibar, which provided a clear model for managing access through a graph of relationships.

Meanwhile, the DevOps team was all-in on AWS. They utilized AWS IAM policies to manage access to cloud resources, including S3 buckets, EC2 instances, and Lambda functions. IAM's JSON-based policy language was deeply integrated into the platform, and although it had its quirks, it provided fine-grained control through built-in support for resource types, actions, and context-aware conditions.

In this chapter, we'll examine each of these policy frameworks—XACML, Rego, OpenFGA, and AWS IAM. Along the way, we'll apply the PARC model introduced in Chapter 7 to understand how each language defines policy, authorizes decisions, represents resources, and uses contextual information. We'll conclude with a comparison that unifies these concepts and compares all of these languages to Cedar.

8.1 XACML: the first policy language standard

Before "policy as code" became a rallying cry for cloud-native developers, authorization policies were often embedded within application logic or enforced through inflexible, product-specific rules. The eXtensible Access Control Markup Language (XACML) (https://docs.oasis-open.org/xacml/3.0/xacml-3.0-core-spec-os-en.html), introduced by OASIS in 2003, was one of the first significant efforts to standardize authorization policy as a separate, declarative language.

XACML not only defined a language for writing policies but also established a comprehensive architecture for how decisions should be evaluated and enforced. The authorization reference architecture introduced in chapter 4 was initially proposed as part of the XACML standard. Recall that this included components like Policy Decision Points (PDPs) and Policy Enforcement Points (PEPs). For a time, especially in government and